University of Minnesota custom Azure course

Session 2 Participant Workbook

Long-Term Archival, Data Movement at Scale, and Hybrid Identity

Azure Storage, Data Protection, and Resilience at Scale

Session: Day 1, Session 2, four hours

Use: Checkpoint discussion, design exercise or lab debrief, discussion prompts, knowledge check

Keep: Your worksheets are the input to the Session 6 roadmap capstone

All scenarios use synthetic data modeled on a large public research university. No real University of Minnesota data is used.

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1. Session map

Day 1, Session 2: Long-Term Archival, Data Movement at Scale, and Hybrid Identity. Four hours including a fifteen-minute break.

Block	Minutes	Activity
Block 1	55	Archival strategy and immutability foundations
Block 2	45	Data movement tooling and hybrid connectivity
Break	15	
Block 3	50	Hybrid identity and access governance, with demonstration
Block 4	75	Hands-on lab and debrief

1.1 What this session covers

Some university datasets carry retention expectations measured in centuries, and much of the data is irreplaceable. This session separates the retention decision (stable for decades, owned by data stewards) from the technology decision (expected to change many times), then builds the technical pattern that makes that separation safe: Archive tier economics, integrity manifests and periodic validation, container-level and version-level WORM, locked versus unlocked policies, and format migration planning.

The middle of the session is the practical plan for moving petabytes: AzCopy, Azure Storage Mover, Azure Data Factory, Azure File Sync, and the next-generation Data Box devices, over ExpressRoute or VPN Gateway, with egress modeled before any design routinely pulls data back on premises. The last half is identity: one identity on premises and in the cloud, Azure Files identity sources, Blob RBAC data roles, ABAC conditions on tags, managed identities, and the end of account keys.

1.2 Have ready

- Your current retention schedule or records policy, even informally: which classes of data must be kept for how long, and who signed it.
- The Session 1 workbook, especially the account decisions for the satellite archive class; the lab builds that account.
- A terminal or Cloud Shell signed in to your lab subscription for Block 4.

1.3 Materials you will receive

- Session 2 deck; demo script `S02_Demo_Script_Century_Proof_Landing_Zone`
- `S02_Lab_Guide_Archival_Landing_Zone` (participants) and `S02_Lab_Solution_Archival_Landing_Zone` (instructor)

- Data movement decision tree and the sample integrity manifest and validation script (Reference Card Pack and the Bicep bundle)
- This workbook (participants) and the facilitator guide (instructor)

2. Block 1 checkpoint: three archival questions

Answer for the university's estate as you know it. Short answers are fine; the room compares them.

Question 1

Who owns the retention decision for each dataset, and how is it recorded so that it survives staff turnover and technology change?

Question 2

How will you prove integrity in 20 years? Name the artifact, where it lives, and how often it is checked.

Question 3

Which datasets justify a locked policy, and which a legal hold? What are the consequences of locking at the wrong scope or interval?

3. Debrief design sprint: your archival landing zone

The lab built one landing zone with synthetic data. In the twenty-minute debrief, each group applies the pattern to the university's estate: choose three real archival datasets or collections (described in your own words), decide their retention class, policy type and lock decision, movement path into Azure, and the identity model that governs who reads them.

Use the lab's results as evidence: how long rehydration took, which refusal you saw, and what the ABAC condition did.

3.1 Steps

1. Record your lab observations in Worksheet 1 (rehydration elapsed time, refusal message, ABAC result).
2. Pick three archival datasets or collections the group knows and complete Worksheet 2.
3. For the largest of the three, complete Worksheet 3: the movement plan and the identity model.
4. Be ready to state one guarantee the platform now gives you that process gave you before.

3.2 Worksheet: lab evidence

Observation	Your result
Time from rehydration request to completion (or elapsed so far)	
First line of the delete refusal	
What the ABAC condition allowed and denied	
Validation result and matched count	

3.3 Worksheet: three archival datasets

Decision	Dataset 1	Dataset 2	Dataset 3
Dataset (your words) and approximate size			
Retention class and years; who owns it			
Policy type (container-level or version-level) and interval			
Lock decision and legal hold (yes or no, why)			
Tier path (Cold floor or Archive) and small-object handling			
Integrity: manifest cadence and where the second copy lives			

3.4 Worksheet: movement and identity for the largest dataset

Decision	Your answer
Volume, timeline, and available bandwidth	
Tool: AzCopy, Storage Mover, Data Factory, File Sync, or Data Box (which device and how many)	
Connectivity: ExpressRoute (peering type) or VPN Gateway; egress expected	
Identity source for file shares (AD DS, Entra Domain Services, Entra Kerberos) or Blob RBAC	
The ABAC condition in words (role, action, attribute, value)	
How keys and SAS are governed (user delegation SAS, shared key disabled)	

3.5 What you present

- Three datasets with retention class, policy, lock decision, and integrity plan.
- One movement and identity plan for the largest dataset.
- One sentence: the guarantee the platform now gives you that process gave you before.

3.6 How your design will be critiqued

Lens	Meets	Exceeds
Reliability	Integrity manifest with two copies and a validation cadence	Validation records kept as audit evidence; migration rule stated
Security	Identity-based access, no keys, ABAC condition stated	Private endpoints over the hybrid link; user delegation SAS only
Cost optimization	Archive versus Cold floor justified by object size and access	Rehydration tests and format migrations budgeted as known recurring costs
Operational excellence	Movement tool chosen from volume, timeline, and network	Data Box count and timeline computed; cutover plan stated
Performance efficiency	Small files bundled before archiving	ExpressRoute peering type matched to private endpoints

4. Discussion prompts

Work through these in your group or with a neighbor during the time the facilitator calls out. Write enough that you can defend the position in the room.

Prompt 1

The retention decision and the technology decision: where does the university make them today, and who would notice if they were mixed up?

Block 1 framing.

Prompt 2

Moving 1.6 PB: compute the timeline over the network you have, then compare with Data Box 525 devices. Where is the crossover?

Block 2. Use the available bandwidth you actually have, not the circuit size.

Prompt 3

Which ABAC condition would replace the most ACLs in your environment, and what process must own the tags for it to be safe?

Block 3.

5. Knowledge check

Six questions, one best answer each. Mark your answer; the facilitator reviews them at the end of the session.

1. What is the maximum interval for a container-level time-based retention policy?

- A. 10 years
- B. 100 years
- C. 146,000 days
- D. Unlimited

2. Which statement about a locked container-level policy is true?

- A. It can be shortened by the account owner

- B. It can be extended up to five times and never shortened or deleted
- C. It can be deleted after 24 hours
- D. It blocks reads

3. Which tool is designed for managed NFS and SMB share migrations at terabyte to petabyte scale over the network?

- A. AzCopy
- B. Azure Storage Mover
- C. Azure Data Box Disk
- D. Azure File Sync

4. Which Data Box devices are current?

- A. Data Box Heavy and Data Box 80 TB
- B. Data Box 120 and Data Box 525
- C. Data Box Edge only
- D. Data Box Gateway only

5. How many identity sources can be enabled on one storage account for Azure Files SMB shares?

- A. One
- B. Two
- C. Three
- D. Unlimited

6. Which SAS type is recommended because it is signed with Microsoft Entra credentials rather than the account key?

- A. Account SAS
- B. Service SAS
- C. User delegation SAS
- D. Stored access policy SAS

6. Takeaways and the capstone

- Archival is a strategy: retention owned by stewards, integrity proven by manifests, immutability declared on the platform, and format migration planned.
- Choose movement tools from volume, timeline, and network; compute the timeline; model egress before any design pulls data back.
- Identity, ABAC, and managed identities replace keys and ACL sprawl; one identity source per Azure Files account; disable shared key access.

- Lock deliberately: test unlocked, validate scope and interval, lock within about a day, and treat the lock as a Security change.

Capstone input from this session

Century-proof archival landing zone with integrity validation.

Next: Day 2 opens with Session 3, data protection in both directions: backup copies, vault-level immutability, and controls that hold against a compromised administrator.

7. Notes

Your notes